THE OHIO STATE UNIVERSITY

Stability of near-merger in the actuation of sound change

Tianyi Ni • Department of Linguistics, The Ohio State University • Speech Prosody 2026

Cantonese T2–T5 • exemplar dynamics • functional load • production–perception asymmetry

QP 2
Poster session-ready • 48 × 36 in

Take-home message. Near-merger need not be a temporary halfway point to merger. In an exemplar model with partially independent production and perception stores, **lexical support** can stabilize a small production residue while perceptual categories collapse or remain substantially weaker.

1. Empirical target: Cantonese T2–T5

Problem. Near-merger exposes a gap between what speakers produce and what listeners can reliably hear: a contrast can persist phonetically even when perceptual categorization has weakened.

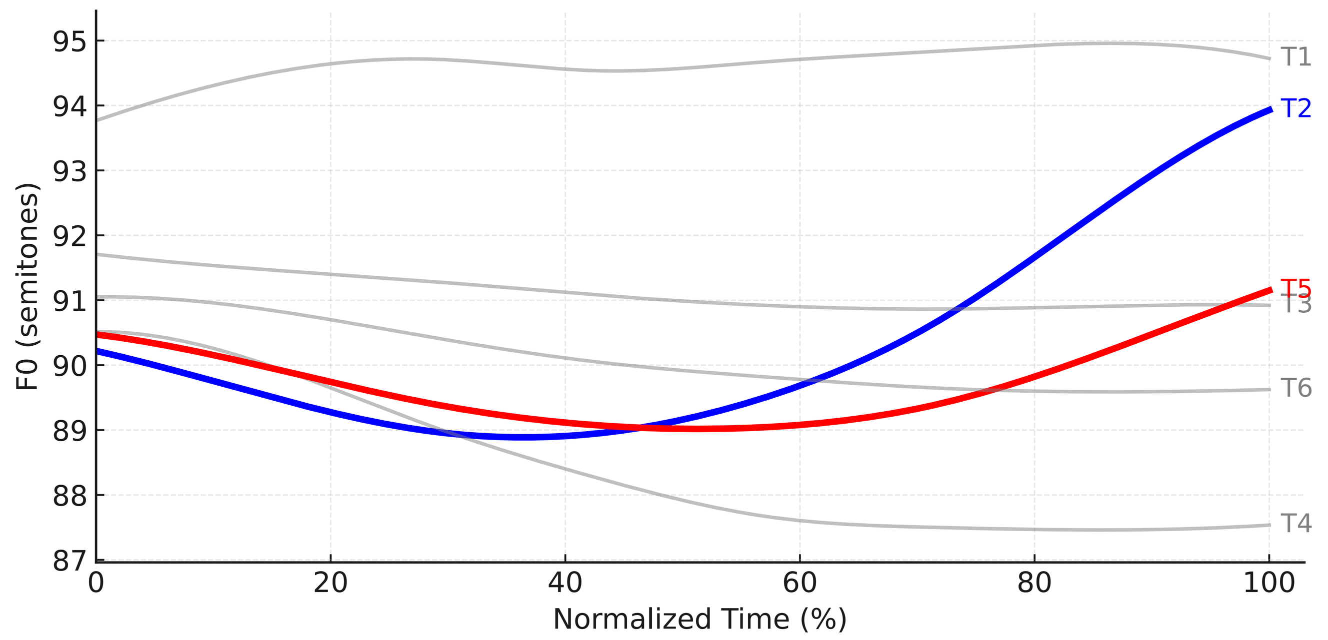


Fig. 1. Six Cantonese citation-tone contours with the two rising tones highlighted. T2 and T5 share contour shape and differ most clearly in offset height and rise magnitude.

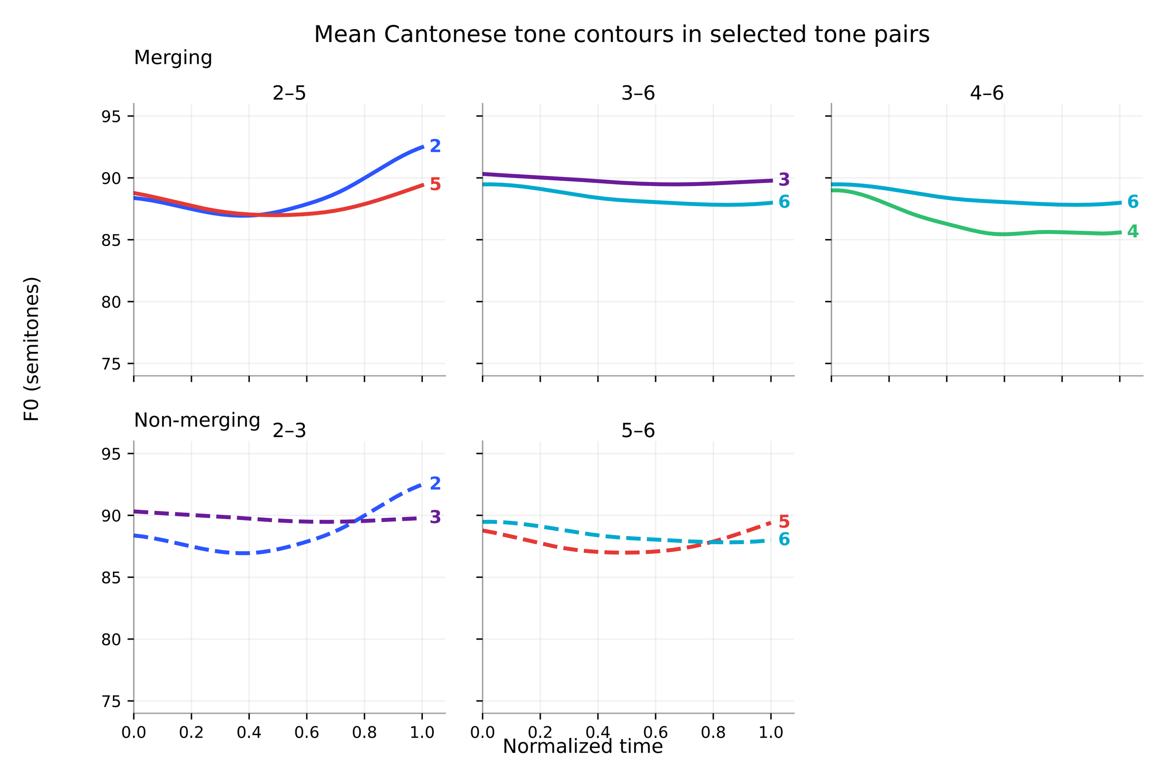


Fig. 2. The broader tone space: reported merger pairs occupy crowded tonal regions and differ by subtle contour or offset-height cues.

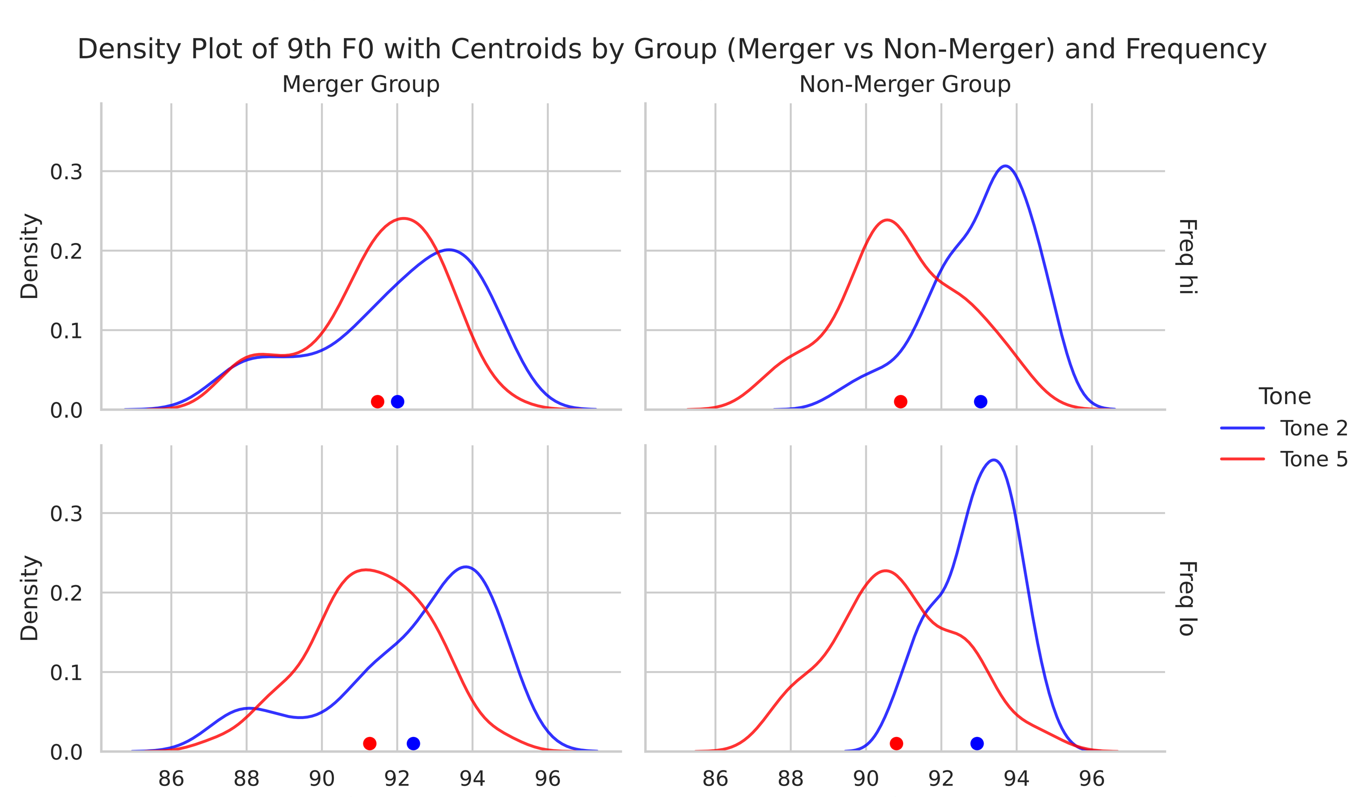


Fig. 3. Density of the 9th normalized F0 point. Non-merging speakers show clearer T2–T5 separation; merger-group productions show stronger overlap in the region that carries the contrast.

Data basis. Monosyllabic Hong Kong Cantonese productions from Mok et al. (2013), initialized from non-merging speakers as the closest pre-merger reference state. Each token contributes ten internal time-normalized F0 measurements in semitones.

Why this contrast is ideal.
- T2–T5 sits in a crowded region of the tone space.
- The contrast is phonetically small but not absent in production.
- Prior work reports slower, less stable perception for confusable Cantonese tone pairs.

2. Hypothesis

Near-merger is a **locally stable equilibrium**: a state generated by the interaction of reduction pressure, lexical contrast pressure, and asymmetric memory updating.

Core prediction. When functional load is intermediate, production remains positive but perception is close to zero; when lexical support is absent, both merge; when lexical support is very high, both retain contrast.

3. Exemplar production–perception loop

The model maintains separate but interacting exemplar stores. Production is lexically recoverable; perception stores ambiguous evidence across categories. This asymmetry lets the two domains diverge over time.

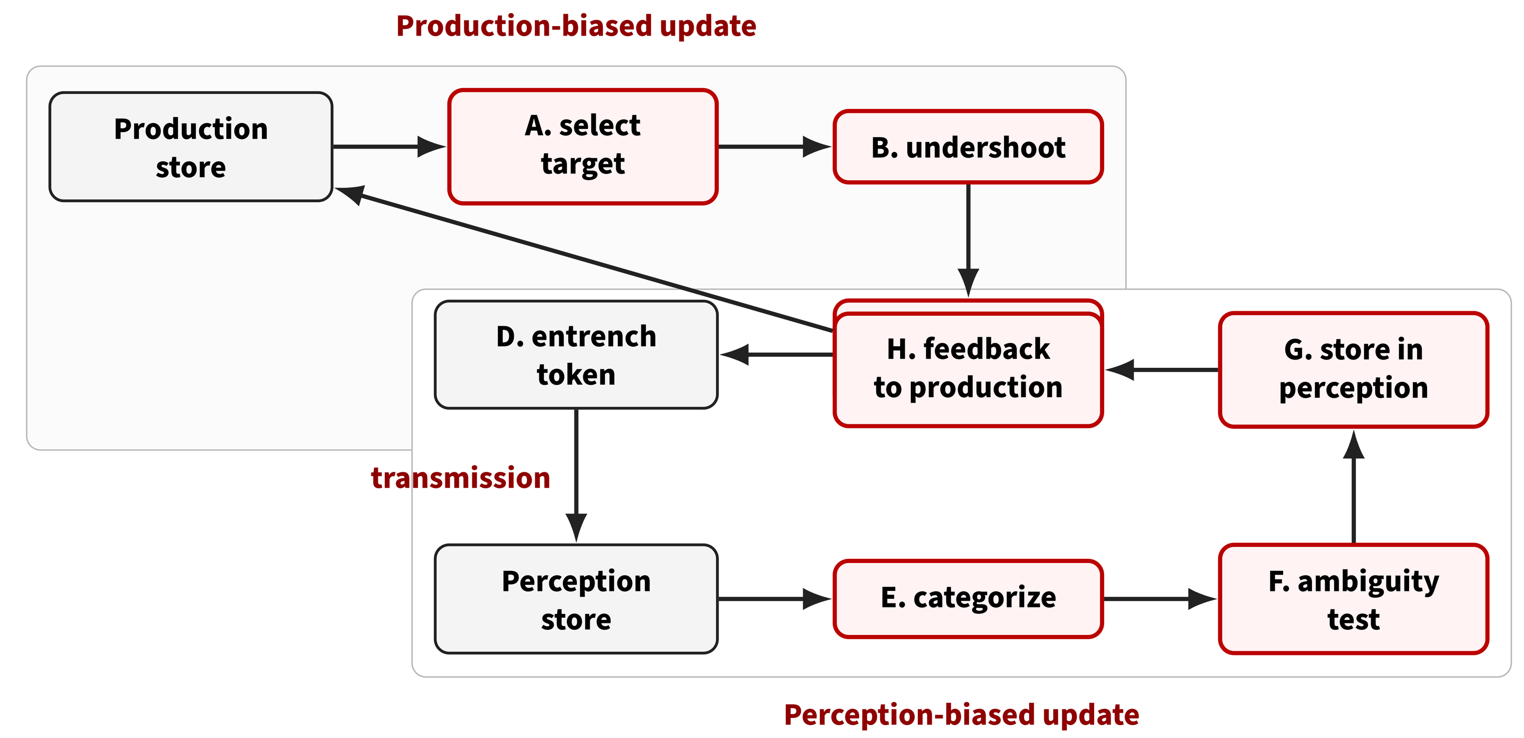


Fig. 4. Closed feedback loop. Ambiguous perceptual evidence can pull categories together, while production storage is corrected through intended lexemes and minimal-pair structure.

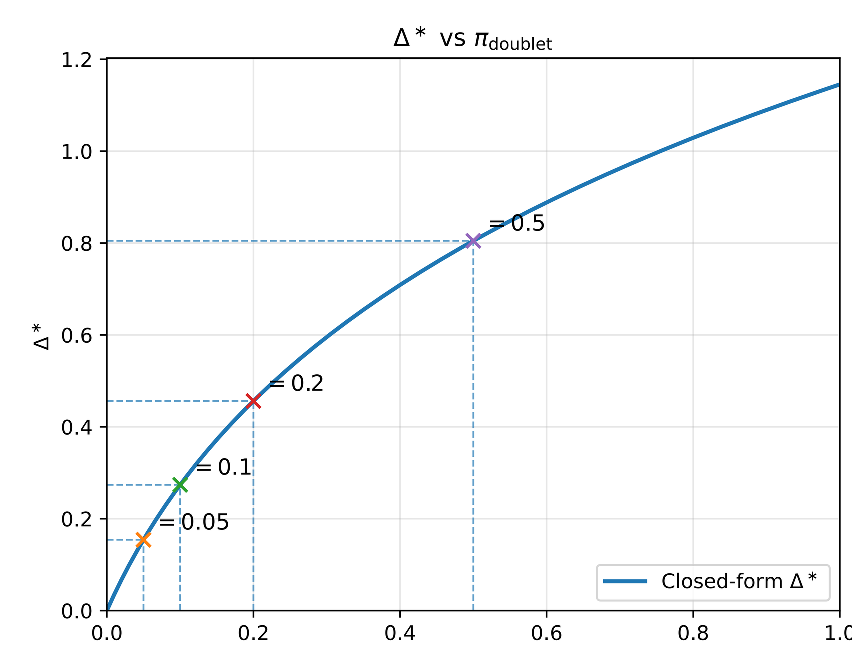
Production update. A sampled target is reduced toward an onset anchor, then doublets receive contrast enhancement away from the competing category:

$$x_c^{\mathrm{und}} = x_{\mathrm{onset}} + (x_c^{\mathrm{target}} - x_{\mathrm{onset}})(1 - e^{-dk_c})$$
$$x_c^{\mathrm{con}} = x_c^{\mathrm{und}} + s_c \beta_{\max} \exp(-|x_c^{\mathrm{und}} - \mu_{\neg c}^{\mathrm{prod}}|).$$

4. Equilibrium logic

Functional load is operationalized as the proportion of minimal-pair competitors, π_{mp}. The closed-form production equilibrium is

$$\Delta^*_{\mathrm{prod}} = A \pi_{mp} e^{-\Delta^*_{\mathrm{prod}}}, \qquad A = \beta_{\max}\left(\frac{1}{\gamma_2} + \frac{1}{\gamma_5}\right).$$



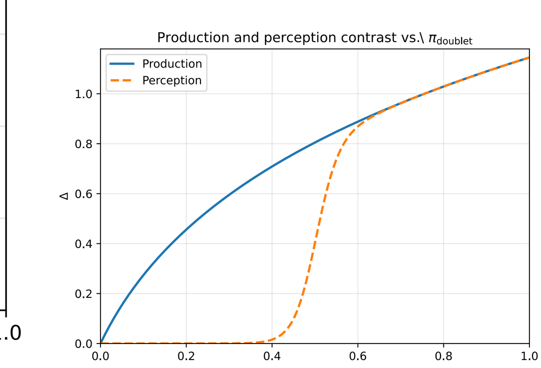


Fig. 5. More lexical support yields greater equilibrium production distance. Fig. 6. The broad region where production is positive but perception remains much smaller is the near-merger regime.

4b. Perceptual ambiguity

Perception is modeled as categorization against the current perceptual exemplar distributions:

$$S_c(x) = \sum_{i \in c} w_i^{\mathrm{perc}} \exp\left[-\frac{(x - x_i^{\mathrm{perc}})^2}{2h^2}\right], \qquad P(c \mid x) = \frac{S_c(x)}{S_2(x) + S_5(x)}.$$

A token is ambiguous when

$$|P(2 \mid x) - P(5 \mid x)| \le \delta.$$

Crucial asymmetry. Ambiguous tokens are stored in both perceptual categories, but production writes back to the intended lexeme. This makes perceptual convergence easier than production convergence.

5. Baseline dynamics: decoupling emerges

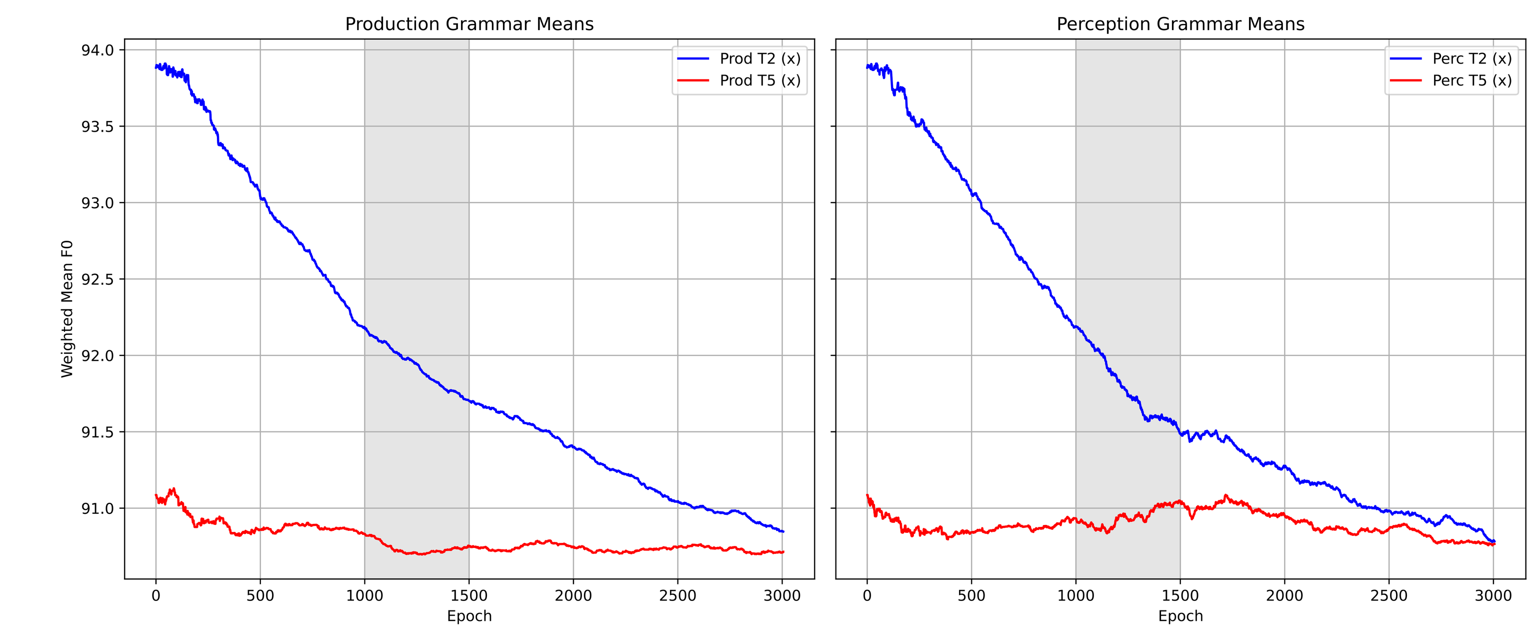


Fig. 7. Production and perception start from the same exemplar distributions. Perception compresses faster, while production retains a small residue of contrast.

Mechanism. Perception stores ambiguous tokens in both categories, accumulating averaging pressure. Production stores tokens in the intended lexeme after lexical recovery, so ambiguity does not erase production separation in the same way.

6. Lexical support determines the outcome

FL	Δ_{prod}	Δ_{perc}
0/7	0.000	0.000
1/7	0.048	0.000
2/7	0.055	0.011
3/7	0.098	0.013
4/7	0.152	0.016
5/7	0.236	0.016
6/7	0.716	0.027
7/7	3.579	3.494

Table 1. Convergent T2–T5 distances in semitones.

Production remains ahead of perception.
At low/intermediate FL, production separation is positive while perceptual distance is approximately zero. Full contrast is recovered only when lexical support is very strong.

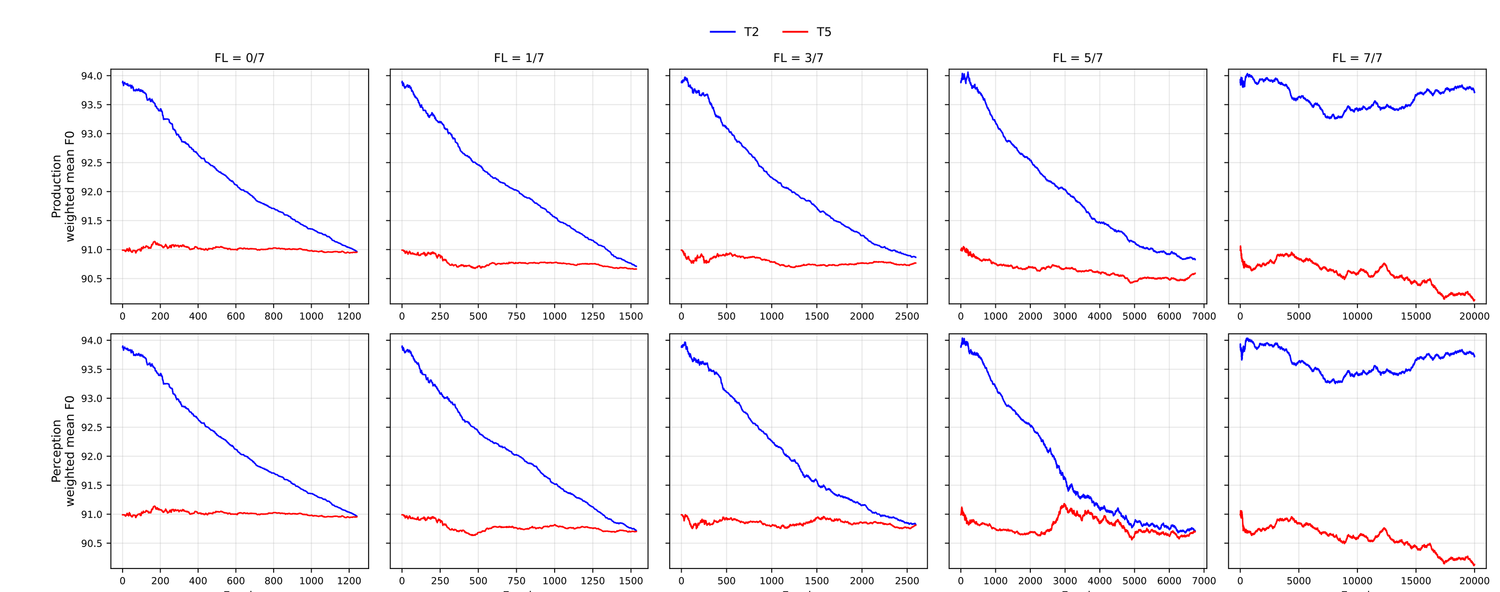


Fig. 8. Trajectories across lexical-support conditions. Low FL converges quickly to merger; higher FL extends the trajectory and preserves progressively larger production separation.

7. Interpretation

Near-merger is not accidental or anomalous. It is an equilibrium configuration made available by the interaction of:
- **phonetic pressure**: undershoot drives categories together;
- **lexical pressure**: minimal-pair competitors preserve contrast;
- **modality asymmetry**: perception can collapse before production.

This reframes functional load from a global survival predictor into a structural force that shapes which stable states a sound system can occupy.

8. Contribution

Theoretical payoff. The model derives stability and change from the same update loop instead of treating near-merger as a failed or incomplete merger.
- It predicts persistent production–perception mismatch.
- It makes functional load a condition on equilibrium size.
- It links the actuation problem to learnable, lexically local dynamics.

Selected references. Bauer & Benedict 1997; Blevins & Wedel 2009; Labov 1994, 2011; Labov, Karen & Miller 1991; Mok, Zuo & Wong 2013; Pierrehumbert 2001; Soo & Babel 2023; Wedel, Kaplan & Jackson 2013; Xu & Wang 2001; Zhang 2019.

Contact: ni.386@buckeyemail.osu.edu